

8.6 NUTRIENT CYCLES

8.6.1 Carbon Cycle

8.6.2 Nitrogen Cycle

8.6.3 Phosphorous Cycle

Organisms need a constant supply of nutrients to grow, reproduce and regulate various body functions. The amount of nutrients, such as carbon, nitrogen, phosphorus, calcium, etc., present in the soil at any given time, is referred to as the **standing state**. It varies in different kinds of ecosystems and also on a seasonal basis.

Nutrients are never lost from the ecosystems. They are recycled again and again, indefinitely. The movement of nutrient elements through the various components of an ecosystem is called '*nutrient cycling*'. Such cycles are called biogeochemical cycles (bio: living organism, geo: rocks, air, water).

Nutrient cycles are of two types:

- (a) gaseous and
- (b) sedimentary.

In a gaseous cycle, elements move through the atmosphere. Main reservoirs are atmosphere and oceans (via evaporation). e.g. carbon cycle and nitrogen cycle.

In a sedimentary cycle, elements move from the earth's crust to water and to sediment. Main reservoirs are the soil and sedimentary rocks. e.g. phosphorous cycle and sulphur cycle.

8.6.1 Carbon Cycle

The element carbon constitutes 49 percent of the dry weight of organisms and is next only to water. Among the total carbon quantity present on the Earth, 71 percent is found dissolved in oceans. This 'oceanic reservoir' regulates the amount of carbon dioxide in the atmosphere. It will be interesting to know that the atmosphere contains only about 1 percent of the total global carbon.

'**Fossil fuels**' also represent reservoirs of carbon. Carbon cycling occurs through atmosphere, ocean and through living and dead organisms. carbon is fixed in the biosphere through photosynthesis. A considerable amount of carbon returns to the atmosphere as CO₂ through respiratory activities of the producers and consumers. Decomposers also contribute substantially to CO₂ pool by their processing of waste materials and dead organic matter of land or oceans. Some amount of the fixed carbon is lost to sediments and removed from circulation. Burning of wood, forest fires combustion of organic matter, fossil fuels, volcanic activity etc., are additional sources for releasing CO₂ into the atmosphere.

Human activities have significantly influenced the carbon cycle. Rapid deforestation and massive burning of fossil fuel for energy and transport have significantly increased the rate of release of carbon dioxide into the atmosphere. (see greenhouse effect at 8.8.4)

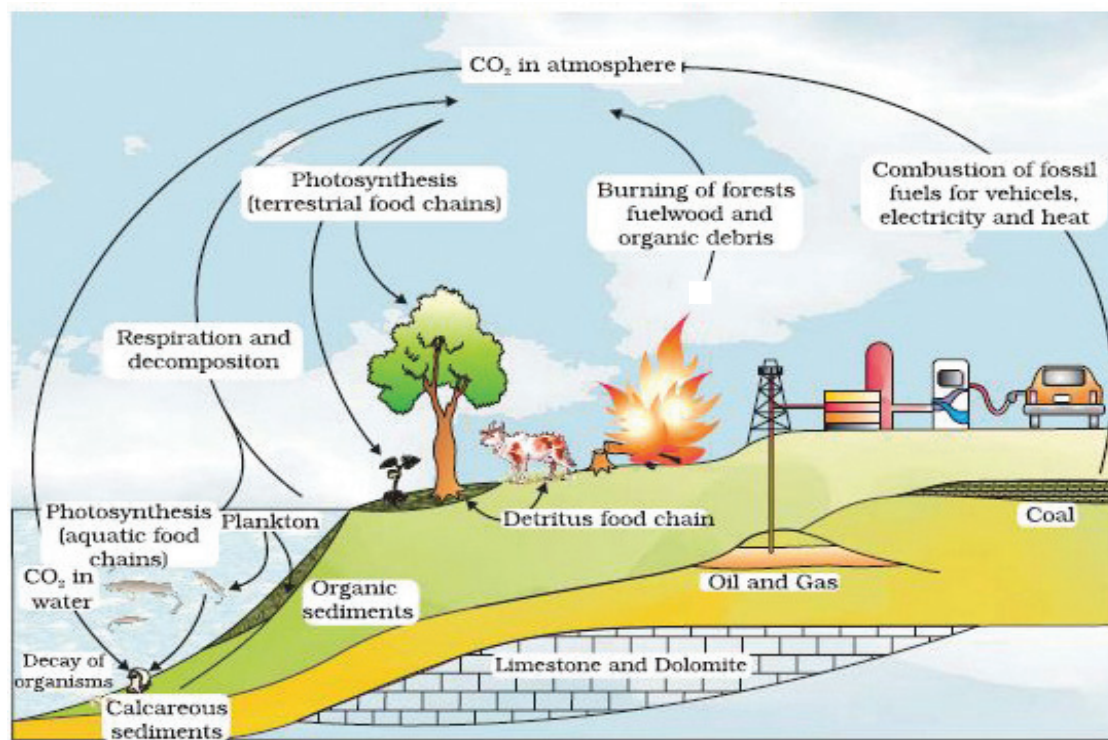


Fig. 8.18 Simplified model of carbon cycle in the biosphere

8.6.2 Nitrogen Cycle

Apart from carbon, hydrogen and oxygen, nitrogen is the most prevalent element in living organisms. Nitrogen is a constituent of amino acids, proteins, hormones, chlorophylls and many of the vitamins. Plants compete with microbes for the limited nitrogen that is available in the soil. Thus nitrogen is a '**limiting nutrient**' for both natural and agricultural ecosystems.

Nitrogen exists in molecular state. The process of conversion of nitrogen (N_2) into nitrites and nitrates is called **nitrogen fixation**. In nature lightening and ultra violet radiation provide enough energy to convert nitrogen to nitrogen oxides NO , NO_2 ,

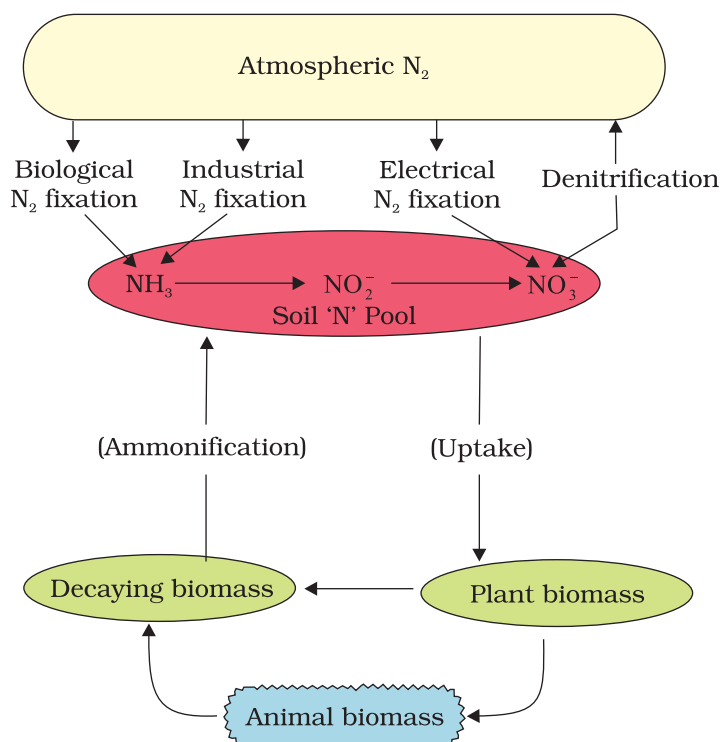
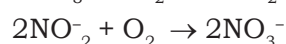


Fig. 8.19 The nitrogen cycle showing relationship between the three main nitrogen pools – atmosphere, soil and biomass

N_2O . Industrial combustions, forest fires, auto-mobile exhausts and power generating stations are also sources for atmospheric nitrogen oxides. Decomposition of organic nitrogen of dead plants and animals into ammonia is called 'ammonification'. Most of the ammonia is converted into nitrites and nitrates by soil bacteria by the following reaction.



Ammonia is first oxidized to nitrite by bacteria such as *Nitrosomonas* and *Nitrosococcus*. Nitrites are further oxidized to 'nitrates' with the help of bacteria such as '*Nitrobacter*'. These steps constitute 'nitrification'. These nitrifying bacteria are *chemoautotrophs*.

The nitrate thus formed is absorbed by plants and is transported to the leaves. In leaves it is reduced to ammonia that finally forms the 'amine' group of amino acids. Nitrates present in the soil are also reduced to nitrogen by the process of 'denitrification'. Denitrification is carried out by bacteria such as *Pseudomonas* and *Thiobacillus*.

8.6.3 Phosphorus Cycle

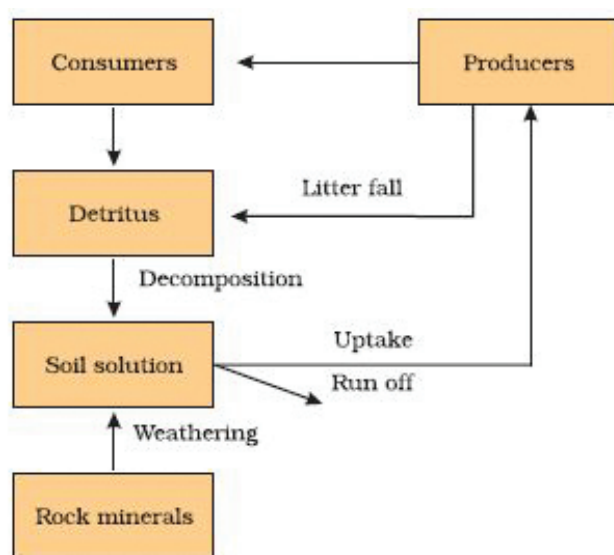


Fig. 8.20 A simplified model of phosphorus cycling in a terrestrial ecosystem

Phosphorus is a major constituent of biological membranes, nucleic acids and cellular energy transfer systems. Many animals also need large quantities of this element to make **shells**, **bones** and **teeth**. The natural reservoir of phosphorus is rock, which contains phosphorus in the form of 'phosphates'. When rocks are '**weathered**', minute amounts of these phosphates dissolve in soil solution and are absorbed by the roots of plants. Herbivores and other animals obtain this element from plants. The waste products and the dead organisms are decomposed by 'phosphate-solubilising bacteria' releasing phosphorus. Unlike carbon cycle, there is no respiratory release of phosphorus into the atmosphere.